

CEO

Cameron Ellis

CEO · ACADEMIC



HIRE

◆ 5

SIGNATURE

Signature: one copy per deck.

Commit:

Play a card that costs 3 Leverage or less by paying Surplus instead of Leverage, at its normal timing.

If you survive the end of round 24, you win regardless of rivals' Surplus. If several companies qualify, use the final standings among them.

the game makes more money the longer you play

Hentges Fellow of Finance and Associate Professor at the University of Iowa. Fellow of the Vaughan Institute of Risk Management.

cellis212.github.io

ALPHA EX MACHINA

NO. 172 / 373

OFFICE LOCATION



OPEN

◆ 3 branch only

Headquarters or Open as a branch.

At the start of your turn, gain 1 Surplus for every 4 Life or Annuity funding sources you control, rounded down.

At the start of your turn, if you control a Life or Annuity funding source and a Private Credit Fund, gain 1 Surplus (once, regardless of copies).

Long lives, long loans, and a very short commute.

ALPHA EX MACHINA

NO. 354 / 373

FUNDING SOURCE

Group Life

LIFE



CAPACITY

◆ 1 Leverage a turn

CARRYING COST

● 1 each turn

▲ Mortality ▼ Longevity

Your maximum hand size is 1 higher.

Life insurance the employer buys for everyone, and nobody remembers signing up for.

ALPHA EX MACHINA

NO. 002 / 373

FUNDING SOURCE

Group Life

LIFE



CAPACITY

◆ 1 Leverage a turn

CARRYING COST

● 1 each turn

▲ Mortality ▼ Longevity

Your maximum hand size is 1 higher.

Life insurance the employer buys for everyone, and nobody remembers signing up for.

ALPHA EX MACHINA

NO. 002 / 373

FUNDING SOURCE

Group Life

LIFE



CAPACITY

◆ 1 Leverage a turn

CARRYING COST

● 1 each turn

▲ Mortality ▼ Longevity

Your maximum hand size is 1 higher.

Life insurance the employer buys for everyone, and nobody remembers signing up for.

ALPHA EX MACHINA

NO. 002 / 373

STAPLE FUNDING SOURCE

Term Life

LIFE



CAPACITY

◆ 1 Leverage a turn

MARGIN

● 1 run start

STAPLE

▲ Mortality ▲ Low Interest Rates ▼ Longevity

Staple: any number per deck.

If uncommitted at the start of your turn, gain its Margin before refreshing.

Coverage for a set term that most customers outlive, which is the business model.

ALPHA EX MACHINA

NO. 003 / 373

STAPLE FUNDING SOURCE

Term Life

LIFE



CAPACITY

◆ 1 Leverage a turn

MARGIN

● 1 run start

STAPLE

▲ Mortality ▲ Low Interest Rates ▼ Longevity

Staple: any number per deck.

If uncommitted at the start of your turn, gain its Margin before refreshing.

Coverage for a set term that most customers outlive, which is the business model.

ALPHA EX MACHINA

NO. 003 / 373

STAPLE FUNDING SOURCE

Term Life

LIFE



CAPACITY

◆ 1 Leverage a turn

MARGIN

● 1 run start

STAPLE

▲ Mortality ▲ Low Interest Rates ▼ Longevity

Staple: any number per deck.

If uncommitted at the start of your turn, gain its Margin before refreshing.

Coverage for a set term that most customers outlive, which is the business model.

ALPHA EX MACHINA

NO. 003 / 373

STAPLE FUNDING SOURCE

Term Life

LIFE



CAPACITY

◆ 1 Leverage a turn

MARGIN

● 1 run start

STAPLE

▲ Mortality ▲ Low Interest Rates ▼ Longevity

Staple: any number per deck.

If uncommitted at the start of your turn, gain its Margin before refreshing.

Coverage for a set term that most customers outlive, which is the business model.

ALPHA EX MACHINA

NO. 003 / 373

Whole Life

LIFE

CAPACITY

1

Leverage a turn

MARGIN

1

run start

STAPLE

▲ Mortality

▲ Low Interest Rates

▼ Longevity

▼ Inflation

Staple: any number per deck.

If uncommitted at the start of your turn, gain its Margin before refreshing.

Lifelong coverage plus a savings account, plus a sales pitch for both.

ALPHA EX MACHINA

NO. 001 / 373

Whole Life

LIFE

CAPACITY

1

Leverage a turn

MARGIN

1

run start

STAPLE

▲ Mortality

▲ Low Interest Rates

▼ Longevity

▼ Inflation

Staple: any number per deck.

If uncommitted at the start of your turn, gain its Margin before refreshing.

Lifelong coverage plus a savings account, plus a sales pitch for both.

ALPHA EX MACHINA

NO. 001 / 373

Index-Linked Annuity

ANNUITY

CAPACITY

1

Leverage a turn

CARRYING COST

1

each turn

▲ Short-Term Equity

▼ Long-Term Equity

Your maximum hand size is 1 higher.

Stock-market upside with a ceiling, downside with a floor, and an actuary in between.

ALPHA EX MACHINA

NO. 008 / 373

Index-Linked Annuity

ANNUITY

CAPACITY

1

Leverage a turn

CARRYING COST

1

each turn

▲ Short-Term Equity

▼ Long-Term Equity

Your maximum hand size is 1 higher.

Stock-market upside with a ceiling, downside with a floor, and an actuary in between.

ALPHA EX MACHINA

NO. 008 / 373

Index-Linked Annuity

ANNUITY

CAPACITY

1

Leverage a turn

CARRYING COST

1

each turn

▲ Short-Term Equity

▼ Long-Term Equity

Your maximum hand size is 1 higher.

Stock-market upside with a ceiling, downside with a floor, and an actuary in between.

ALPHA EX MACHINA

NO. 008 / 373

Fixed Annuity

ANNUITY

CAPACITY

1

Leverage a turn

MARGIN

0

run start

STAPLE

▲ Longevity

▲ Low Interest Rates

▼ Mortality

Staple: any number per deck.

Guaranteed checks for the customer; interest-rate nightmares for you.

ALPHA EX MACHINA

NO. 010 / 373

Fixed Annuity

ANNUITY

CAPACITY

1

Leverage a turn

MARGIN

0

run start

STAPLE

▲ Longevity

▲ Low Interest Rates

▼ Mortality

Staple: any number per deck.

Guaranteed checks for the customer; interest-rate nightmares for you.

ALPHA EX MACHINA

NO. 010 / 373

Fixed Annuity

ANNUITY

CAPACITY

1

Leverage a turn

MARGIN

0

run start

STAPLE

▲ Longevity

▲ Low Interest Rates

▼ Mortality

Staple: any number per deck.

Guaranteed checks for the customer; interest-rate nightmares for you.

ALPHA EX MACHINA

NO. 010 / 373

Fixed Annuity

ANNUITY

CAPACITY

1

Leverage a turn

MARGIN

0

run start

STAPLE

▲ Longevity

▲ Low Interest Rates

▼ Mortality

Staple: any number per deck.

Guaranteed checks for the customer; interest-rate nightmares for you.

ALPHA EX MACHINA

NO. 010 / 373

INVESTMENT

Corn Farms

CROP



COST

◆ 2

YIELD

● 1 each turn

SALE

1 any time

▲ Volatility

▼ Inflation

Yield +1 while you have an office in Des Moines.

Destroyed if a Nat Cat Stress Test hits you.

Planted in April, priced in Chicago, and paid for by the weather.

ALPHA EX MACHINA

NO. 104 / 373

INVESTMENT

Corn Farms

CROP



COST

◆ 2

YIELD

● 1 each turn

SALE

1 any time

▲ Volatility

▼ Inflation

Yield +1 while you have an office in Des Moines.

Destroyed if a Nat Cat Stress Test hits you.

Planted in April, priced in Chicago, and paid for by the weather.

ALPHA EX MACHINA

NO. 104 / 373

INVESTMENT

Soy Farms

CROP



COST

◆ 2

YIELD

● 1 each turn

SALE

1 any time

▲ Volatility

▼ Inflation

Crop rotation. Yield +1 while this is paired with a Corn Farms you control, one to one.

Destroyed if a Nat Cat Stress Test hits you.

Corn one year and soybeans the next, because the nitrogen comes free even if the price doesn't.

ALPHA EX MACHINA

NO. 105 / 373

INVESTMENT

Soy Farms

CROP



COST

◆ 2

YIELD

● 1 each turn

SALE

1 any time

▲ Volatility

▼ Inflation

Crop rotation. Yield +1 while this is paired with a Corn Farms you control, one to one.

Destroyed if a Nat Cat Stress Test hits you.

Corn one year and soybeans the next, because the nitrogen comes free even if the price doesn't.

ALPHA EX MACHINA

NO. 105 / 373

INVESTMENT

Private Credit Fund

ILLIQUID



COST

◆ 5

YIELD

● 3 each turn

ILLIQUID

▲ Short-Term Equity

▼ Inflation

Illiquid: can't be liquidated.

Destroyed if Dash for Cash, Northern Rock Bank Run, or Great Depression hits you.

Marked to model, quarterly.

ALPHA EX MACHINA

NO. 071 / 373

INVESTMENT

Private Credit Fund

ILLIQUID



COST

◆ 5

YIELD

● 3 each turn

ILLIQUID

▲ Short-Term Equity

▼ Inflation

Illiquid: can't be liquidated.

Destroyed if Dash for Cash, Northern Rock Bank Run, or Great Depression hits you.

Marked to model, quarterly.

ALPHA EX MACHINA

NO. 071 / 373

INVESTMENT

Private Equity

ILLIQUID



COST

◆ 3

ILLIQUID

▲ Regulation

Illiquid: can't be liquidated.

Your maximum hand size is 1 higher.

Destroyed if a Liquidity Stress Test hits you.

No distributions yet, which the fund calls the J-curve.

ALPHA EX MACHINA

NO. 052 / 373

INVESTMENT

Private Equity

ILLIQUID



COST

◆ 3

ILLIQUID

▲ Regulation

Illiquid: can't be liquidated.

Your maximum hand size is 1 higher.

Destroyed if a Liquidity Stress Test hits you.

No distributions yet, which the fund calls the J-curve.

ALPHA EX MACHINA

NO. 052 / 373

RISK MANAGEMENT

Risk Committee Veto

RESPONSE



COST

◆ 3

RESPONSE

Response: play while a Stress Test is being run.

Stop a Stress Test that hits you. This cannot Stop an Act of God.

The minutes will reflect that we said no.

ALPHA EX MACHINA

NO. 165 / 373

RISK MANAGEMENT

Risk Committee Veto

RESPONSE

COST
♦ 3

RESPONSE

Response: play while a Stress Test is being run.

Stop a Stress Test that hits you. This cannot Stop an Act of God.

The minutes will reflect that we said no.

ALPHA EX MACHINA NO. 165 / 373

RISK MANAGEMENT

Risk Committee Veto

RESPONSE

COST
♦ 3

RESPONSE

Response: play while a Stress Test is being run.

Stop a Stress Test that hits you. This cannot Stop an Act of God.

The minutes will reflect that we said no.

ALPHA EX MACHINA NO. 165 / 373

RISK MANAGEMENT

Crisis Communications

RESPONSE

COST
♦ 2

RESPONSE

Response: play while a Stress Test is being run.

Stop a Reputation Stress Test that hits you.

"We take this matter extremely seriously," said the statement drafted in advance.

ALPHA EX MACHINA NO. 129 / 373

RISK MANAGEMENT

Crisis Communications

RESPONSE

COST
♦ 2

RESPONSE

Response: play while a Stress Test is being run.

Stop a Reputation Stress Test that hits you.

"We take this matter extremely seriously," said the statement drafted in advance.

ALPHA EX MACHINA NO. 129 / 373

SIGNATURE RISK MANAGEMENT

Industry Lobbying

RESPONSE

COST
♦ 3

SIGNATURE

RESPONSE

Signature: one copy per deck; Response: play while a Stress Test is being run.

Stop a Regulation Stress Test that hits you.

The trade association would like to submit four hundred pages of comments.

ALPHA EX MACHINA NO. 128 / 373

RISK MANAGEMENT

Incident Response Retainer

RESPONSE

COST
♦ 2

RESPONSE

Response: play while a Stress Test is being run.

Stop a Cyber Stress Test that hits you.

The forensics team is on a plane before the ransom note finishes loading.

ALPHA EX MACHINA NO. 130 / 373

SIGNATURE RISK MANAGEMENT

Everything Is Fine

RESPONSE

COST
♦ 2

DARK

SIGNATURE

RESPONSE

Dark: Stress Tests can charge for it on your board or in your discard pile; Signature: one copy per deck; Response: play while a Stress Test is being run.

Stop a Liquidity or Reputation Stress Test that hits you.

The chairman was calm, confident, and on live television. Two of those were true.

ALPHA EX MACHINA NO. 149 / 373

TALENT

Executive Assistant

FRONT LINE

HIRE
♦ 2

Commit: Uncommit a different non-funding card you control. Use this ability only once between the starts of your turns.

Found an opening in your calendar and put you in it.

ALPHA EX MACHINA NO. 171 / 373

TALENT

Executive Assistant

FRONT LINE

HIRE
♦ 2

Commit: Uncommit a different non-funding card you control. Use this ability only once between the starts of your turns.

Found an opening in your calendar and put you in it.

ALPHA EX MACHINA NO. 171 / 373

TALENT

Corporate Spy

FRONT LINE

HIRE

◆ 2

DARK

Dark: Stress Tests can charge for it on your board or in your discard pile.

Each opponent's maximum hand size is 1 lower.

Badge says Facilities, laptop says otherwise.

ALPHA EX MACHINA

NO. 214 / 373

TALENT

Risk Management Professor

ACADEMIC

HIRE

◆ 2

When you hire this employee, search your deck for a Risk Management card, show it, and put it in your hand. Then shuffle your deck.

Wrote the textbook your CRO still keeps on the desk.

ALPHA EX MACHINA

NO. 211 / 373

TOOL

Delegation Dashboard

SOFTWARE

PRICE

◆ 3

▲ Cyber

Commit: Uncommit a different non-funding card you control. Use this ability only once between the starts of your turns.

That meeting could have been somebody else's meeting.

ALPHA EX MACHINA

NO. 359 / 373

SIGNATURE TOOL

Generative AI Pilot

SOFTWARE

PRICE

◆ 2

CARRYING COST

● 1 each turn

SIGNATURE

AI

▲ Cyber

Signature: one copy per deck; AI: AI Bubble Bursts destroys it.

Your maximum hand size is 1 higher.

Summarizes every report in the building, including three that don't exist.

ALPHA EX MACHINA

NO. 368 / 373

SURPRISE INSPECTION

Whistleblower

CORPORATE

COST

◆ 3

HITS

Target opponent

Target opponent loses 6 Surplus per Dark card in their discard pile.

After losses, each company hit discards every investment named Shell Company, Feeder Fund, Affiliate Loan, or Guaranteed Returns Fund it controls.

She kept copies of everything, including the email telling her not to.

ALPHA EX MACHINA

NO. 297 / 373

SURPRISE INSPECTION

Reinsurer Insolvency

MARKET

COST

◆ 1

HITS

Target opponent

Target opponent has one Risk Management card to discard, chosen by the runner.

The treaty was airtight. The reinsurer was not.

ALPHA EX MACHINA

NO. 334 / 373

SIGNATURE STRESS TEST

Dash for Cash

LIQUIDITY · AS-IF REPLAY, 2020

HITS

Every company

SIGNATURE

▲ Liquidity

▲ Short-Term Equity

▲ Credit

▲ Volatility

Signature: one copy per deck.

Every company loses 3 Surplus per Liquidity exposure, 3 per Short-Term Equity exposure, 3 per Credit exposure, and 2 per Volatility exposure.

After losses, each company hit discards every investment named Private Credit Fund it controls.

In March 2020, investors sold almost everything, even U.S. Treasuries, to raise cash, until the Federal Reserve stepped in.

ALPHA EX MACHINA

NO. 269 / 373

SIGNATURE STRESS TEST

Flash Crash

MARKET · AS-IF REPLAY, 2010

HITS

Every company

SIGNATURE

ACT OF GOD

▲ Short-Term Equity

▲ Volatility

Signature: one copy per deck; Act of God: can't be stopped.

Every company loses 6 Surplus per Short-Term Equity exposure and 2 per Volatility exposure.

After losses, each company hit discards every investment named Meme Stock it controls.

On May 6, 2010, a burst of automated selling briefly wiped about a trillion dollars off the value of U.S. stocks. The Dow fell almost 1,000 points in minutes, then mostly came back.

ALPHA EX MACHINA

NO. 283 / 373

SIGNATURE STRESS TEST

Roundup Verdict

LITIGATION · AS-IF REPLAY, 2018

HITS

Every company

SIGNATURE

▲ Litigation

Signature: one copy per deck.

Every company loses 8 Surplus per Litigation exposure.

In August 2018, a San Francisco jury ordered Monsanto to pay \$289 million to a groundskeeper who blamed its weedkiller, Roundup, for his cancer.

ALPHA EX MACHINA

NO. 282 / 373

Asbestos Claims

LITIGATION



HITS

Every company

SIGNATURE

ACT OF GOD

DEVELOPING EVENT

▲ Litigation

Signature: one copy per deck; Act of God: can't be stopped;
Developing Event: all stages this round, each with a response window.

Stage 1. Every company loses 4 Surplus per Litigation exposure.

Stage 2. Every company loses 4 Surplus per Litigation exposure.

Stage 3. Every company loses 4 Surplus per Litigation exposure.

The exposure was decades old, and the claims kept coming for decades more.

ALPHA EX MACHINA

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1906 San Francisco Earthquake

NAT CAT · AS-IF REPLAY, 1906



HITS

Every company

SIGNATURE

ACT OF GOD

▲ Nat Cat: Earthquake

▲ Property Value

Signature: one copy per deck; Act of God: can't be stopped.

Every company loses 12 Surplus per Nat Cat: Earthquake exposure and 4 per Property Value exposure.

Surprise trust fall.

ALPHA EX MACHINA

NO. 251 / 373

Hurricane Harvey

NAT CAT · AS-IF REPLAY, 2017



HITS

Every company

SIGNATURE

ACT OF GOD

▲ Nat Cat: Flood

Signature: one copy per deck; Act of God: can't be stopped.

Every company loses 10 Surplus per Nat Cat: Flood exposure.

In August 2017, Hurricane Harvey stalled over southeast Texas for days and dropped up to 60 inches of rain, flooding much of Houston.

ALPHA EX MACHINA

NO. 237 / 373

1927 Mississippi River Flood

NAT CAT · AS-IF REPLAY, 1927



HITS

Every company

SIGNATURE

ACT OF GOD

▲ Nat Cat: Flood

Signature: one copy per deck; Act of God: can't be stopped.

Every company loses 10 Surplus per Nat Cat: Flood exposure.

In 1927 the Mississippi River broke through its levees in more than 100 places, flooding 27,000 square miles and displacing hundreds of thousands of people.

ALPHA EX MACHINA

NO. 252 / 373

Cancer Cured

LONGEVITY



HITS

Every company

SIGNATURE

▲ Longevity

Signature: one copy per deck.

Every company loses 12 Surplus per Longevity exposure.

Champagne in the lab, followed by a very quiet meeting in annuity reserving.

ALPHA EX MACHINA

NO. 246 / 373

Volcker Shock

RATES · AS-IF REPLAY, 1979



HITS

Every company

SIGNATURE

▲ Short-Term Rates

▲ Liquidity

▲ Volatility

Signature: one copy per deck.

Every company loses 6 Surplus per Short-Term Rates exposure, 2 per Liquidity exposure, and 2 per Volatility exposure.

In October 1979 the Federal Reserve let short-term rates climb toward 20% to break inflation, and the savings and loans that had funded 30-year mortgages with passbook savings never recovered.

ALPHA EX MACHINA

NO. 301 / 373

Opioid Epidemic

MORTALITY



HITS

Every company

SIGNATURE

▲ Mortality

▲ Litigation

▲ Health Price

Signature: one copy per deck.

Every company loses 4 Surplus per Mortality exposure, 4 per Litigation exposure, and 4 per Health Price exposure.

Brought to you by Purdue Pharma.

ALPHA EX MACHINA

NO. 243 / 373

Zero-Commission Price War

MARKET · AS-IF REPLAY, 2019



HITS

Every company

SIGNATURE

Signature: one copy per deck.

Every company loses 2 Surplus. Then each company commits 2 of its available funding sources.

On October 1, 2019, Charles Schwab cut its online stock trading commissions to zero, and its biggest rivals matched within days.

ALPHA EX MACHINA

NO. 286 / 373

2003 European Heat Wave

MORTALITY · AS-IF REPLAY, 2003



HITS

Every company

SIGNATURE

ACT OF GOD

▲ Mortality

▲ Health Price

Signature: one copy per deck; Act of God: can't be stopped.

Every company loses 4 Surplus per Mortality exposure and 4 per Health Price exposure.

In the summer of 2003, a heat wave killed more than 70,000 people across Europe, most of them elderly. The forecast called it a week in advance, and the emergency rooms filled anyway.

ALPHA EX MACHINA

NO. 233 / 373

AWS S3 Outage

CYBER · AS-IF REPLAY, 2017



HITS

Every company

SIGNATURE

▲ Cyber

Signature: one copy per deck.

Every company loses 6 Surplus per Cyber exposure, plus 3 more. Then each company discards a card.

On February 28, 2017, a mistyped command knocked Amazon's S3 storage offline for about four hours, and Amazon's status page could not show it, because it ran on S3.

ALPHA EX MACHINA

NO. 284 / 373

Iowa Derecho

NAT CAT · AS-IF REPLAY, 2020



HITS

Every company

SIGNATURE

ACT OF GOD

▲ Nat Cat

Signature: one copy per deck; Act of God: can't be stopped.

Every company loses 2 Surplus per Nat Cat exposure.

On August 10, 2020, a derecho crossed Iowa with winds up to 140 mph, flattening corn, grain bins, and most of the trees in Cedar Rapids.

ALPHA EX MACHINA

NO. 339 / 373

Cape Wind Abandoned

MARKET · AS-IF REPLAY, 2017



HITS

Every company

SIGNATURE

▲ Long-Term Equity

Signature: one copy per deck.

Every company loses 10 Surplus per Long-Term Equity exposure.

In December 2017, after sixteen years of fights with its neighbors, the company behind Cape Wind, planned as America's first offshore wind farm, gave up. Nothing got built.

ALPHA EX MACHINA

NO. 256 / 373

Klarna's Down Round

MARKET · AS-IF REPLAY, 2022



HITS

Every company

SIGNATURE

▲ Credit

Signature: one copy per deck.

Every company loses 5 Surplus per Credit exposure.

In July 2022, Klarna, Europe's biggest buy now, pay later lender, raised money at a valuation 85% below its peak a year earlier. Four easy payments, then the fifth, the sixth, and the collections calls.

ALPHA EX MACHINA

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